Qwen3.5: Towards Native Multimodal Agents

Reformatted from the official release materials

Qwen Team (original release)
This LaTeX version: autogenerated from public materials

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Abstract

Qwen3.5 is a native multimodal mixture-of-experts (MoE) foundation model designed for agentic workloads that require reasoning over language, images, and interactive tool use. According to the official release materials, Qwen3.5 combines (i) early-fusion multimodal pretraining, (ii) an efficient hybrid architecture based on Gated Delta Networks with sparse MoE routing, and (iii) reinforcement learning (RL) scaled to large numbers of agent environments. This document reorganizes the public release content into a self-contained LaTeX paper, while preserving the original tables and figures.[1, 2]

Figure 1: Qwen3.5 logo used in the official release materials.[2]

1 Introduction

Recent "agentic" applications increasingly require models to (a) interpret multimodal inputs, (b) plan multi-step actions, and (c) interact with external tools under long contexts. The Qwen3.5 release positions the model family as an attempt to unify these requirements in a native multimodal foundation model.[1] This paper-like write-up focuses on the first open-weight release, `Qwen3.5-397B-A17B`, and summarizes its architecture, capabilities, and benchmark results as reported by the authors.[2]

2 Model overview

The released checkpoint is described as a causal language model with a vision encoder and a sparse MoE backbone.[2] Key properties are summarized in Table 1.

3 Design highlights (from the release materials)

The official release emphasizes the following themes: [1, 2]

Figure 2: Release banner screenshot from public release reporting.[3]

Table 1: Selected configuration details for `Qwen3.5-397B-A17B` as reported in the official model card.[2]

Item	Value
Total parameters	397B
Active parameters	17B
Layers	60
Hidden dimension	4096
Vocabulary size	248,320 (padded)
MoE experts	512 (10 routed + 1 shared activated)
Context length	262,144 native; extensible to 1,010,000
Backbone	$15 \times [3 \times$ (Gated DeltaNet $\rightarrow$ MoE) $+ 1 \times$ (Gated Attention $\rightarrow$ MoE)]

- **Unified vision–language foundation.** Early-fusion training on large multimodal corpora aims to align visual understanding with language reasoning.
- **Efficient hybrid architecture.** Gated Delta Networks and sparse MoE routing are used to improve throughput and reduce inference overhead.
- **Scalable RL generalization.** RL is reported to be scaled across large collections of tool-using agent environments with progressively more complex task distributions.
- **Broad multilingual coverage.** The release materials claim support for 201 languages and dialects.

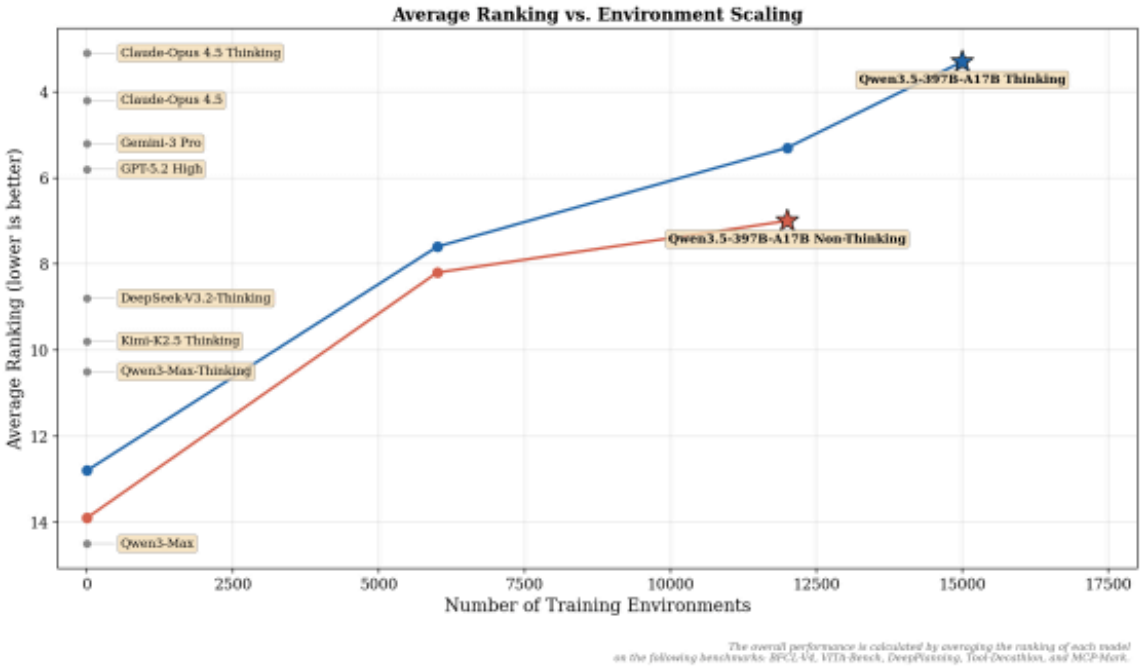

Figure 3: Example figure on RL environment scaling reproduced in public release reporting.[3]

4 Benchmarks

4.1 Summary figures

Figure 4 reproduces the benchmark summary figure from the official release repository.[2]

4.2 Language benchmarks

Table 2 lists the detailed language results extracted from the official model card (HTML table preserved and converted into LaTeX).[2]

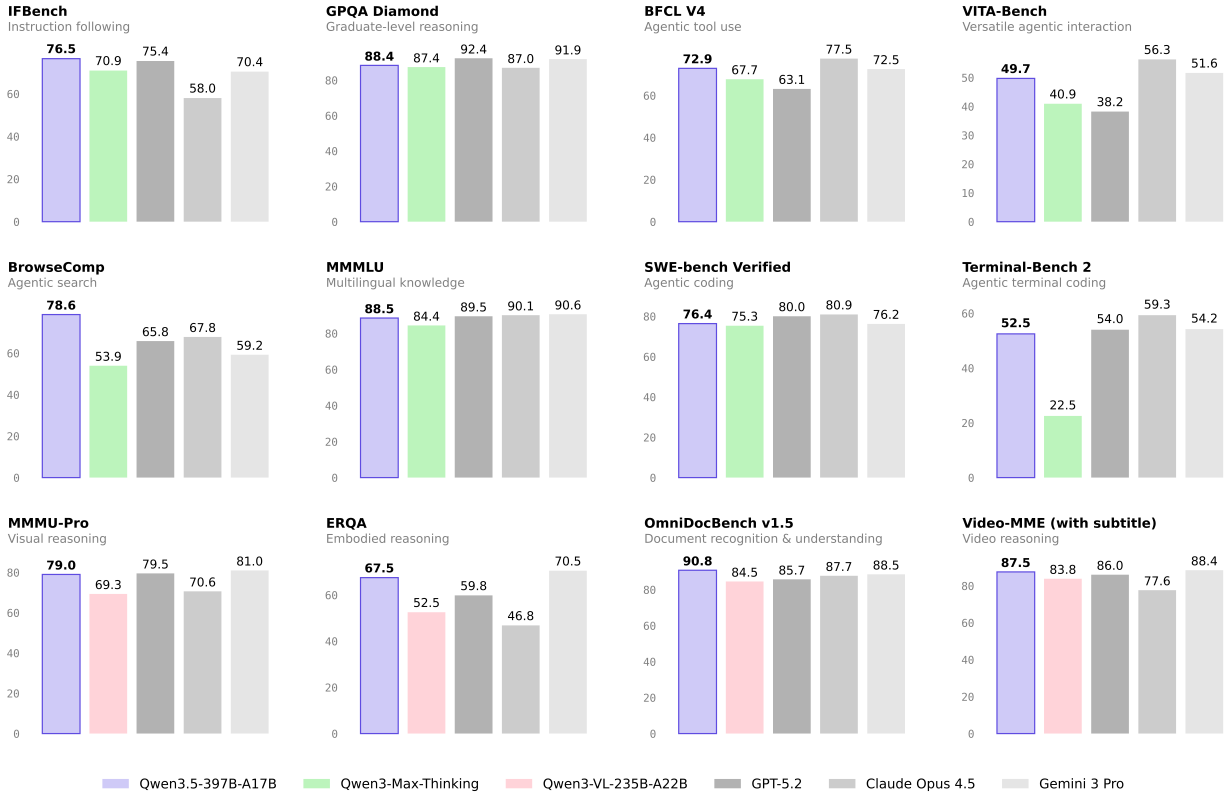

Figure 4: Benchmark score overview for `Qwen3.5-397B-A17B` as provided in the official release materials.[2]

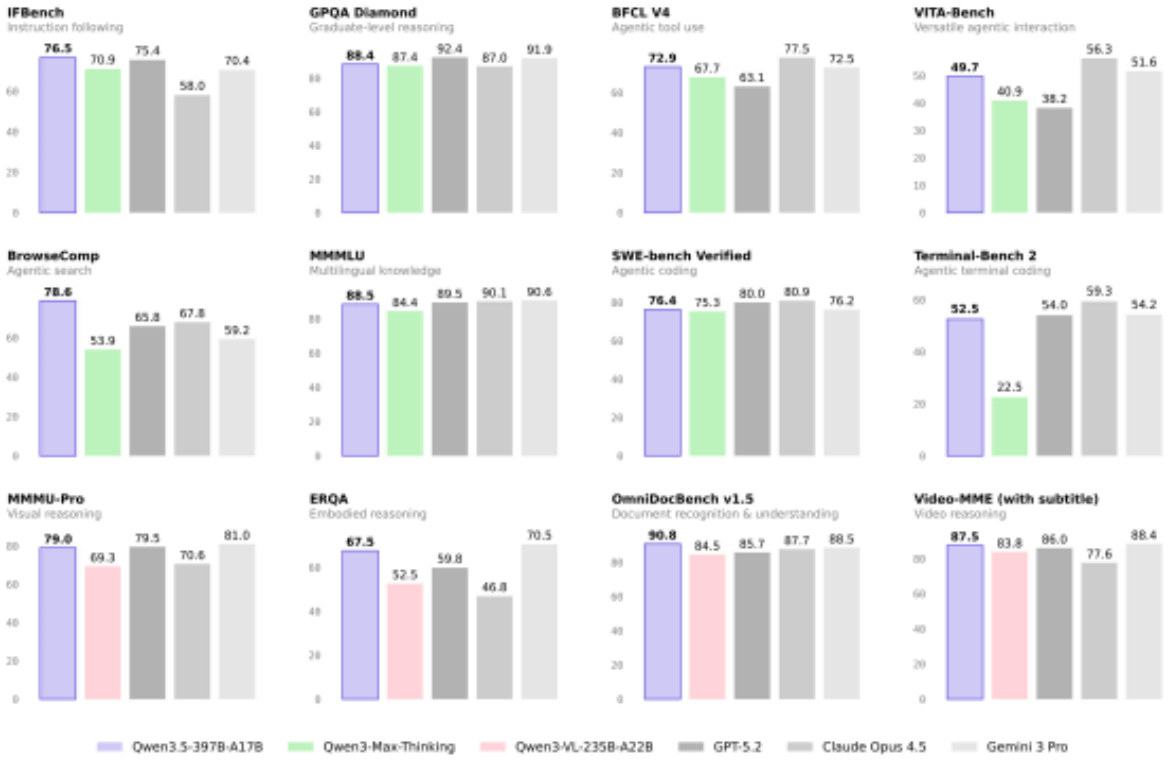

Figure 5: Additional benchmark panel screenshot reproduced in public release reporting.[3]

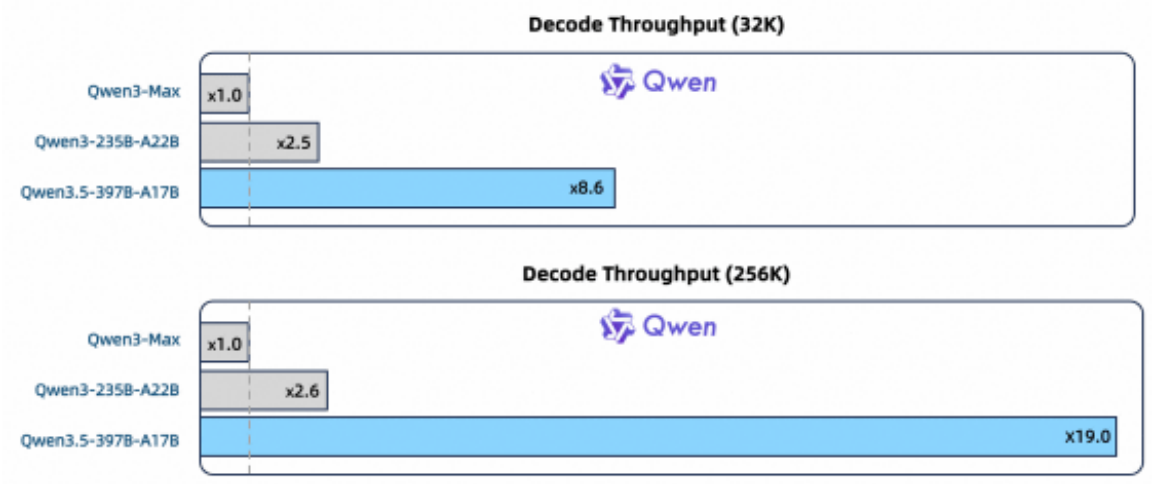

Figure 6: Decode throughput chart reproduced in public release reporting.[3]

Table 2: Language benchmark results reported in the Qwen3.5 model card.

Metric	GPT5.2	Claude 4.5 Opus	Gemini-3 Pro	Qwen3-Max-Thinking	K2.5-1T-A32B	Qwen3.5-397B-A17B
Knowledge						
MMLU-Pro	87.4	89.5	89.8	85.7	87.1	87.8
MMLU-Redux	95.0	95.6	95.9	92.8	94.5	94.9
SuperGPQA	67.9	70.6	74.0	67.3	69.2	70.4
C-Eval	90.5	92.2	93.4	93.7	94.0	93.0
Instruction Following						
IFEval	94.8	90.9	93.5	93.4	93.9	92.6
IFBench	75.4	58.0	70.4	70.9	70.2	76.5
MultiChallenge	57.9	54.2	64.2	63.3	62.7	67.6
Long Context						
AA-LCR	72.7	74.0	70.7	68.7	70.0	68.7
LongBench v2	54.5	64.4	68.2	60.6	61.0	63.2
STEM						
GPQA	92.4	87.0	91.9	87.4	87.6	88.4
HLE	35.5	30.8	37.5	30.2	30.1	28.7
HLE-Verified[1]	43.3	38.8	48	37.6	–	37.6
Reasoning						
LiveCodeBench v6	87.7	84.8	90.7	85.9	85.0	83.6
HMMT Feb 25	99.4	92.9	97.3	98.0	95.4	94.8
HMMT Nov 25	100	93.3	93.3	94.7	91.1	92.7
IMOAnswerBench	86.3	84.0	83.3	83.9	81.8	80.9
AIME26	96.7	93.3	90.6	93.3	93.3	91.3
General Agent						
BFCL-V4	63.1	77.5	72.5	67.7	68.3	72.9
TAU2-Bench	87.1	91.6	85.4	84.6	77.0	86.7
VITA-Bench	38.2	56.3	51.6	40.9	41.9	49.7
DeepPlanning	44.6	33.9	23.3	28.7	14.5	34.3
Tool Decathlon	43.8	43.5	36.4	18.8	27.8	38.3
MCP-Mark	57.5	42.3	53.9	33.5	29.5	46.1
Search Agent[3]						
HLE w/ tool	45.5	43.4	45.8	49.8	50.2	48.3
BrowseComp	65.8	67.8	59.2	53.9	–/74.9	69.0/78.6
BrowseComp-zh	76.1	62.4	66.8	60.9	–	70.3
WideSearch	76.8	76.4	68.0	57.9	72.7	74.0
Seal-0	45.0	47.7	45.5	46.9	57.4	46.9
Multilingualism						
MMMLU	89.5	90.1	90.6	84.4	86.0	88.5
MMLU-ProX	83.7	85.7	87.7	78.5	82.3	84.7
NOVA-63	54.6	56.7	56.7	54.2	56.0	59.1
INCLUDE	87.5	86.2	90.5	82.3	83.3	85.6
Global PIQA	90.9	91.6	93.2	86.0	89.3	89.8
PolyMATH	62.5	79.0	81.6	64.7	43.1	73.3
WMT24++	78.8	79.7	80.7	77.6	77.6	78.9
MAXIFE	88.4	79.2	87.5	84.0	72.8	88.2
Coding Agent						
SWE-bench Verified	80.0	80.9	76.2	75.3	76.8	76.4
SWE-bench Multilingual	72.0	77.5	65.0	66.7	73.0	69.3
SecCodeBench	68.7	68.6	62.4	57.5	61.3	68.3
Terminal Bench 2	54.0	59.3	54.2	22.5	50.8	52.5

4.3 Vision–language benchmarks

Table 3 lists the detailed vision–language results extracted from the official model card (HTML table preserved and converted into LaTeX).[2]

Table 3: Vision-language benchmark results reported in the Qwen3.5 model card.

Metric	GPT5.2	Claude 4.5 Opus	Gemini-3 Pro	Qwen3-VL-235B-A22B	K2.5-1T-A32B	Qwen3.5-397B-A17B
STEM and Puzzle						
MMMU	86.7	80.7	87.2	80.6	84.3	85.0
MMMU-Pro	79.5	70.6	81.0	69.3	78.5	79.0
MathVision	83.0	74.3	86.6	74.6	84.2	88.6
Mathvista(mini)	83.1	80.0	87.9	85.8	90.1	90.3
We-Math	79.0	70.0	86.9	74.8	84.7	87.9
DynaMath	86.8	79.7	85.1	82.8	84.4	86.3
ZEROBench	9	3	10	4	9	12
ZEROBench_sub	33.2	28.4	39.0	28.4	33.5	41.0
BabyVision	34.4	14.2	49.7	22.2	36.5	52.3/43.3
General VQA						
RealWorldQA	83.3	77.0	83.3	81.3	81.0	83.9
MMStar	77.1	73.2	83.1	78.7	80.5	83.8
HallusionBench	65.2	64.1	68.6	66.7	69.8	71.4
MMBenchEN-DEV-v1.1	88.2	89.2	93.7	89.7	94.2	93.7
SimpleVQA	55.8	65.7	73.2	61.3	71.2	67.1
Text Recognition and Document Understanding						
OmniDocBench1.5	85.7	87.7	88.5	84.5	88.8	90.8
CharXiv(RQ)	82.1	68.5	81.4	66.1	77.5	80.8
MMLongBench-Doc	–	61.9	60.5	56.2	58.5	61.5
CC-OCR	70.3	76.9	79.0	81.5	79.7	82.0
AI2D_TEST	92.2	87.7	94.1	89.2	90.8	93.9
OCRBench	80.7	85.8	90.4	87.5	92.3	93.1
Spatial Intelligence						
ERQA	59.8	46.8	70.5	52.5	–	67.5
CountBench	91.9	90.6	97.3	93.7	94.1	97.2
RefCOCO(avg)	–	–	84.1	91.1	87.8	92.3
ODInW13	–	–	46.3	43.2	–	47.0
EmbSpatialBench	81.3	75.7	61.2	84.3	77.4	84.5
RefSpatialBench	–	–	65.5	69.9	–	73.6
LingoQA	68.8	78.8	72.8	66.8	68.2	81.6
V*	75.9	67.0	88.0	85.9	77.0	95.8/91.1
Hypersim	–	–	–	11.0	–	12.5
SUNRGBD	–	–	–	34.9	–	38.3
Nuscene	–	–	–	13.9	–	16.0
Video Understanding						
VideoMME(w sub.)	86	77.6	88.4	83.8	87.4	87.5
VideoMME(w/o sub.)	85.8	81.4	87.7	79.0	83.2	83.7
VideoMMMU	85.9	84.4	87.6	80.0	86.6	84.7
MLVU (M-Avg)	85.6	81.7	83.0	83.8	85.0	86.7
MVBench	78.1	67.2	74.1	75.2	73.5	77.6
LVBench	73.7	57.3	76.2	63.6	75.9	75.5
MMVU	80.8	77.3	77.5	71.1	80.4	75.4
Visual Agent						
ScreenSpot Pro	–	45.7	72.7	62.0	–	65.6

Metric	GPT5.2	Claude 4.5 Opus	Gemini-3 Pro	Qwen3-VL-235B-A22B	K2.5-1T-A32B	Qwen3.5-397B-A17B
OSWorld-Verified	38.2	66.3	–	38.1	63.3	62.2
AndroidWorld	–	–	–	63.7	–	66.8
Medical VQA						
SLAKE	76.9	76.4	81.3	54.7	81.6	79.9
PMC-VQA	58.9	59.9	62.3	41.2	63.3	64.2
MedXpertQA-MM	73.3	63.6	76.0	47.6	65.3	70.0

5 Usage and deployment notes

The release materials provide example commands for serving the model locally via multiple frameworks.[2] Below we reproduce representative snippets.

5.1 Hugging Face Transformers (serving)

```
transformers serve --port 8000 --continuous-batching
# OpenAI-compatible endpoint:
# http://localhost:8000/v1
```

5.2 SGLang

```
python -m sglang.launch_server \
  --model-path Qwen/Qwen3.5-397B-A17B \
  --port 8000 --tp-size 8 \
  --context-length 262144 \
  --reasoning-parser qwen3
```

5.3 vLLM

```
vllm serve Qwen/Qwen3.5-397B-A17B \
  --port 8000 \
  --tensor-parallel-size 8 \
  --max-model-len 262144 \
  --reasoning-parser qwen3
```

References

[1] Qwen Team. Qwen3.5: Towards Native Multimodal Agents. February 2026. https://qwen.ai/blog?id=qwen3.5.

[2] Qwen Team. Qwen3.5-397B-A17B Model Card. 2026. https://huggingface.co/Qwen/Qwen3.5-397B-A17B.

[3] GIGAZINE. Open source release of Qwen3.5-397B-A17B and benchmark summaries (includes screenshots of the release materials). February 2026. https://gigazine.net/news/20260217-qwen3-5-397b-a17b/.